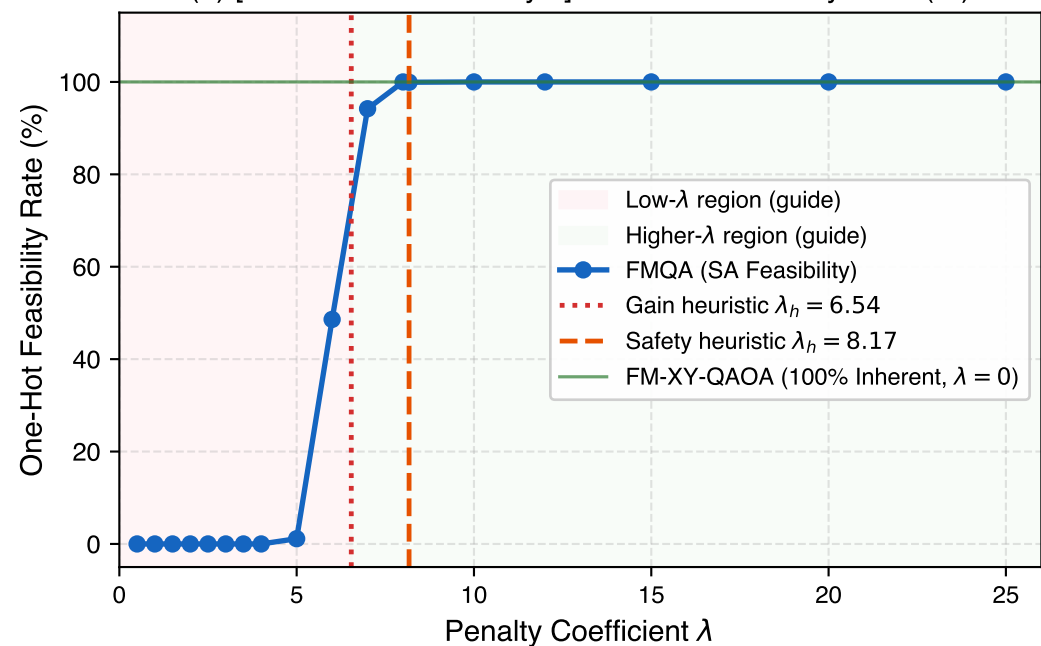
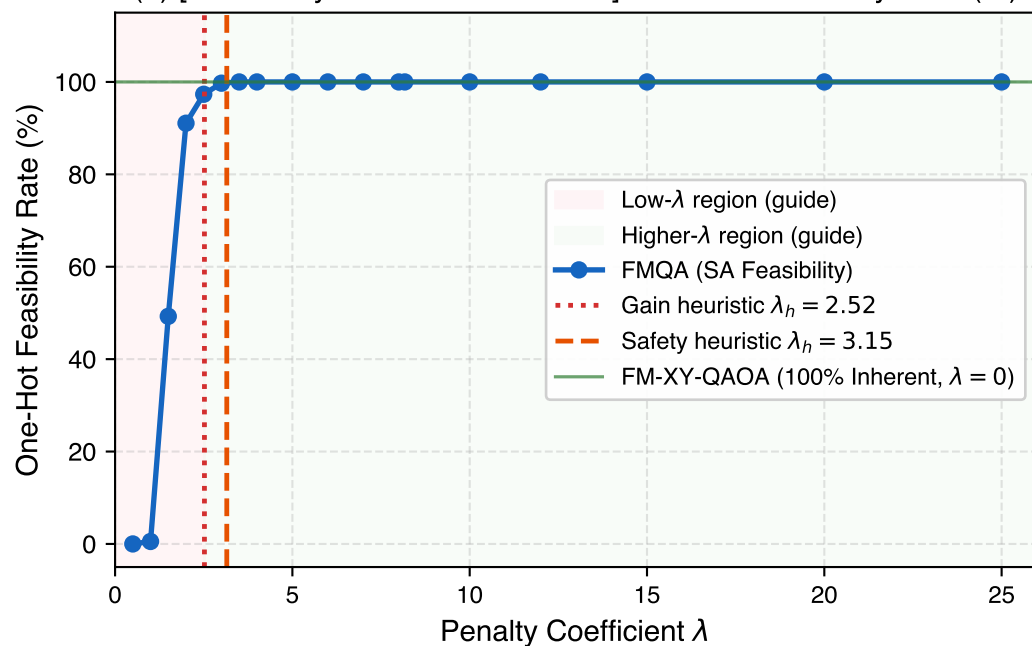


FMQA Penalty Sensitivity Analysis: Penalty Coefficient λ vs Optimization Performance (Comparison against Subspace-Preserving FM-XY-QAOA Baseline)

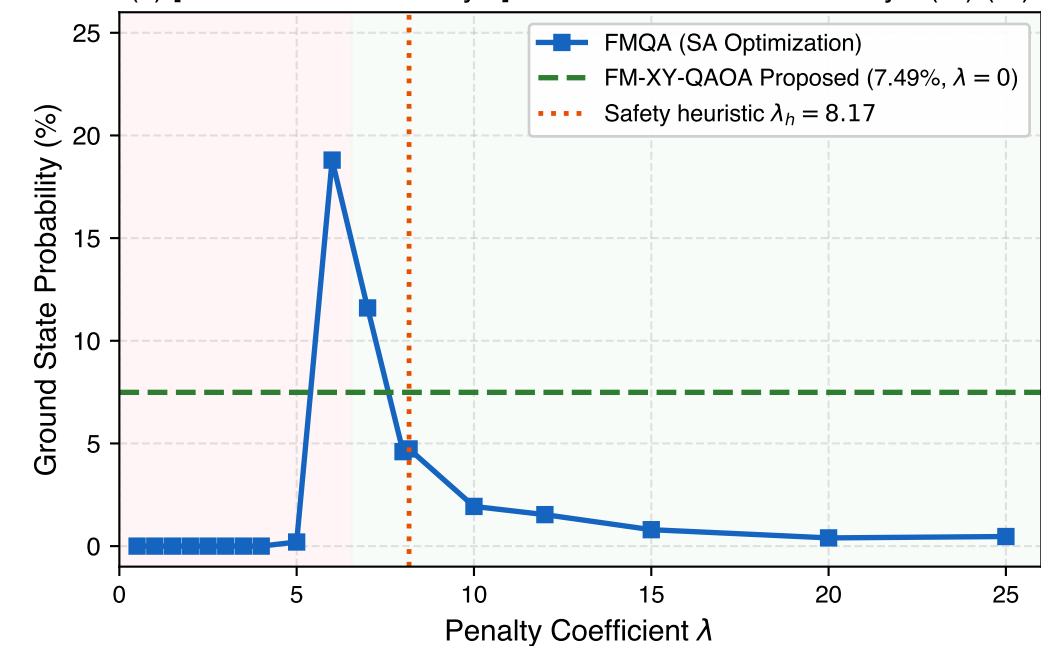
(a) [BB-1 Materials Catalyst] One-Hot Feasibility Rate (%)



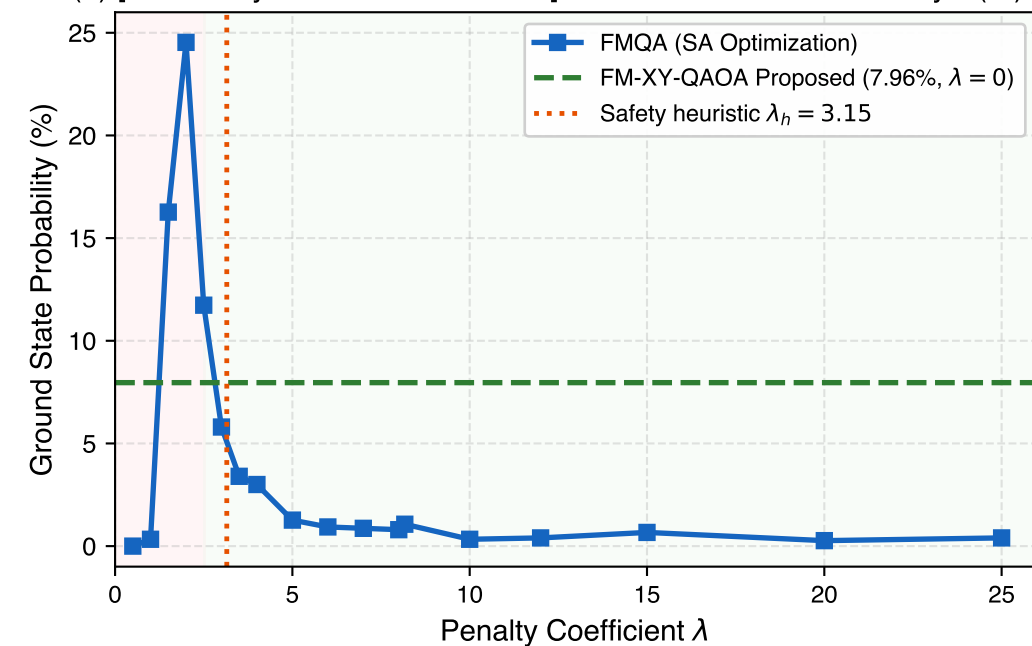
(b) [BB-2 Fully-Connected Random] One-Hot Feasibility Rate (%)



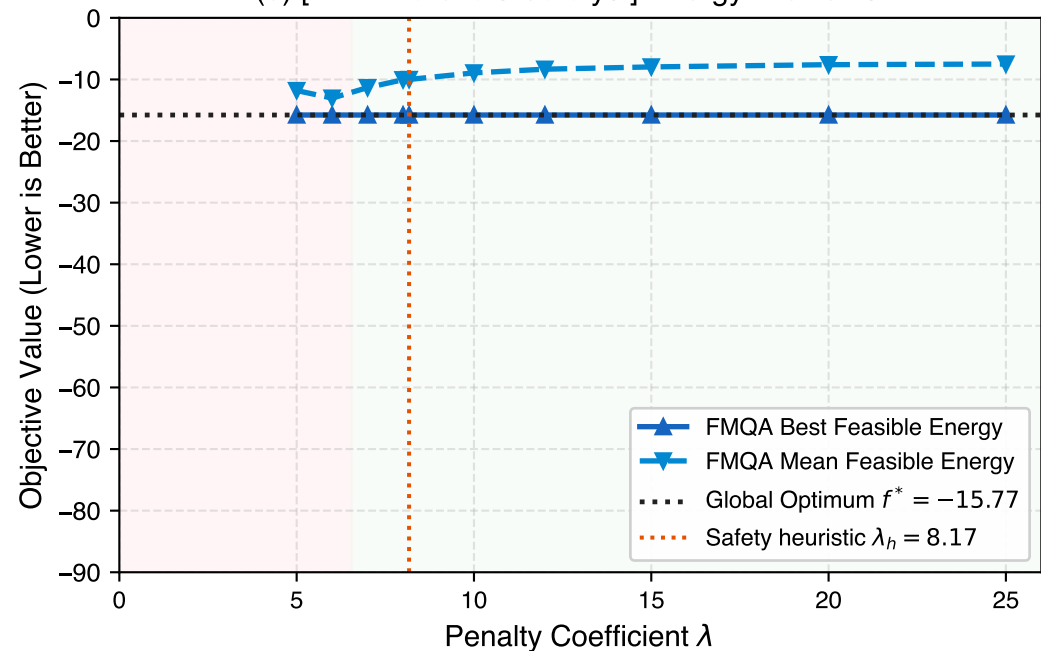
(c) [BB-1 Materials Catalyst] Ground State Hit Probability $P(x^*)$ (%)



(d) [BB-2 Fully-Connected Random] Ground State Hit Probability $P(x^*)$ (%)



(e) [BB-1 Materials Catalyst] Energy Profile vs λ



(f) [BB-2 Fully-Connected Random] Energy Profile vs λ

